Local behavior of Hodge structures at infinity

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Abstract

This article is an expanded version of a letter to D. Morrison, inspired by his article [3]. I give a description of "very degenerating" variations of Hodge structures, which is well adapted to the variations arising in the mirror fairy tale. I also try to ascertain how "motivic" this description is.

0. Introduction

In [3], D. Morrison describes some of the results of [1] in a Hodge theoretic language. One ingredient is the variation of Hodge structures defined by a family of Calabi–Yau manifolds near a point (in a completed parameter space) of "maximal degeneracy". We interpret "maximal degeneracy" as leading to the appearance of variations of mixed Hodge structures of a special type (Hodge-Tate), and give a description of such mixed variations in terms of invertible holomorphic functions. The logarithmic derivatives of those functions are what appears in Morrison's paper.

I thank E. Cattani for a very helpful letter directing me to parts of [2] I had missed, and the referee for suggesting many improvements.

1 1. Notations

For V a polarized variation of Hodge structures of weight w on a complex variety S, we use the following notations.

$V_{\mathbb{Z}}$: underlying local system of free abelian groups. For $R = \mathbb{Q}, \mathbb{R}, \mathbb{C}, \mathscr{O}$ (sheaf of holomorphic functions), $V_R := V_{\mathbb{Z}} \otimes_{\mathbb{Z}} R$.

$\mathbb{Z}(k)$: $(2\pi i)^k\mathbb{Z}$, viewed as a Hodge structure of type $(-k,-k)$. Also: the corresponding constant variation.

$\psi : V_{\mathbb{Z}} \otimes V_{\mathbb{Z}} \to \mathbb{Z}(-w)$: the polarization form, symmetric for w even and alternating for w odd.

At each point $s \in S$, the Hodge structure of the fiber V_s of V at s is given by a decomposition

$$V_{s,\mathbb{C}} = \bigoplus_{p+q=w} V_s^{p,q}$$

with $V_s^{q,p}$ the complex conjugate of $V_s^{p,q}$. The Weil operator C: multiplication by i^{p-q} on $V_s^{p,q}$, is real and the form

$$(2\pi i)^w \psi(Cv, \bar{v})$$

is hermitian symmetric and positive definite.

The holomorphic vector bundle with integrable connection having $V_{\mathbb{C}}$ as sheaf of horizontal sections is identified with its sheaf $V_{\mathscr{O}}$ of holomorphic sections. We denote

∇: its connection.

F: the Hodge filtration. At each point, $F^p = \bigoplus_{a \geq p} V_s^{a,b}$. It is a holomorphic filtration and $\nabla F^p \subset \Omega^1 \otimes F^{p-1}$ (Griffiths transversality).

Typical example

Let $(X_s)_{s\in S}$ be a family of nonsingular projective varieties parametrized by S, i.e. a projective and smooth map $f: X \to S$ (with $X_s = f^{-1}(s)$). Take

$$V_{\mathbb{Z}} := R^w f_* \mathbb{Z} \quad \text{mod torsion}.$$

It is the local system of the $H^w(X_s)$/torsion and $V_s^{p,q} = H^{p,q}(X_s)$.

For X_s purely of dimension n, and $w = n$, on the primitive part of $H^n(X_s)$ (the kernel of the cup-product with the first Chern class of the given ample line bundle), a polarization form is given by

$$\frac{1}{(2\pi i)^n}(-1)^{n(n-1)/2}(2\pi i)^w \int_{X_s} v \cup w.$$

The construction of a polarization form on the whole of the cohomology is by reduction to that case, applied to linear sections of X_s.

2 2. Asymptotics

Let D be the open unit disc, and $D^* := D - \{0\}$. We take $S = D^{*n}$ ($n \geq 1$) and recall known results on the behavior of $V_{\mathbb{R}}$ for $s \to 0$. To the extent possible, our references will be to the survey paper [2].

2.1 2.1. Monodromy

Let T_j be the monodromy of $V_{\mathbb{Z}}$ around $s_j = 0$. The T_j commute and are known to be quasi-unipotent. For simplicity, we assume they are even unipotent.

We define

$$N_j := -\log(T_j)$$

(note the minus sign). Sign convention, explained for $n = 1$: T is the effect of pushing an element of $(V_{\mathbb{Z}})_{s_0}$ horizontally along the path $\exp(2\pi i u)s_0$ ($0 \leq u \leq 1$).

2.2 2.2. Canonical extension

For any complex local system $\mathscr{H}_{\mathbb{C}}$ on D^{*n}, with unipotent monodromy, we continue to write $\mathscr{H}_{\mathscr{O}}$ for the canonical extension of the vector bundle $\mathscr{H}_{\mathscr{O}}$ to D^n. It is characterized by the property that, in a local basis near 0, the matrix of 1-forms defining the connection has logarithmic poles along the $s_j = 0$, with nilpotent residues.

Consider on $\mathscr{H}_{\mathscr{O}}$ the new connection

$$\nabla^0 = \nabla - \frac{1}{2\pi i}\sum_j N_j \frac{ds_j}{s_j}. \tag{2.2.1}$$

The horizontal sections for ∇^0 are the sections of the form $\exp(\sum \log s_j \cdot N_j/2\pi i)h$ for h ∇-horizontal. The connection ∇^0 has no monodromy and turns $\mathscr{H}_{\mathscr{O}}$ into a constant bundle. We will say *constant* to mean horizontal for ∇^0. The canonical extension of $\mathscr{H}_{\mathscr{O}}$ is obtained by extending it as a constant bundle.

The construction of ∇^0 depends on the choice of coordinates near 0, while the canonical extension does not. In fact, the constant vector bundle attached to $\mathscr{H}_{\mathscr{O}}$ should be viewed as living on the tangent space T_0 of D^n at 0. Local coordinates s_j (with s_i/s_j invertible) define (a) an isomorphism φ between a neighborhood of 0 in D^n and a neighborhood of 0 in T_0, (b) a constantification of $\mathscr{H}_{\mathscr{O}}$.

If we use φ to transplant the local system $\mathscr{H}_{\mathbb{C}}$ and the constantification of $\mathscr{H}_{\mathscr{O}}$ to T_0, the result is independent of the local coordinates used.

If a filtration F of the canonical extension $\mathscr{H}_{\mathscr{O}}$ is given, the corresponding nilpotent orbit is the constant filtration of $\mathscr{H}_{\mathscr{O}}$ which coincides with F at 0. Again, to be coordinate free, it should live on T_0.

2.3 2.3. Nilpotent orbit

For a polarized variation V on D^{*n}, the nilpotent orbit theorem states that

(a) The Hodge filtration F extends to the canonical extension $V_{\mathscr{O}}$ as a filtration (still denoted F) by locally direct factors ([2] 2.1 (i)),

(b) The corresponding nilpotent orbit is again, in a neighborhood of 0, a variation of Hodge structures of weight w polarized by ψ ([2] 2.1 (ii)).

Conversely, let $(V_{\mathbb{Z}}, F^{\mathrm{nilp}})$ be a polarized variation of Hodge structures on D^{*n} which is a nilpotent orbit: the filtration F^{nilp} is assumed constant (for ∇^0, see (2.2)). Let F be a new filtration of $V_{\mathscr{O}}$ on D^n, agreeing with F^{nilp} at $s = 0$, and obeying transversality:

$$\nabla F^p \subset \Omega^1 \otimes F^{p-1}.$$

Assume further that $\psi(F^p, F^{w-p+1}) = 0$ for all p. Then $(V_{\mathbb{Z}}, F)$ is, near 0, a variation of Hodge structures on D^{*n}, polarized by ψ. See [2], 2.8.

2.4 2.4. Weight filtration

There exists on $V_{\mathbb{Q}}$ a unique increasing filtration W, the weight filtration, such that $N_j W_k \subset W_{k-2}$ and that for any positive linear combination $N = \sum \lambda_j N_j$ ($\lambda_j > 0$), N^k induces an isomorphism from $\mathrm{Gr}^W_{w+k}(V_{\mathbb{R}})$ to $\mathrm{Gr}^W_{w-k}(V_{\mathbb{R}})$. Further, $(V_{\mathbb{Z}}, W, F^{\mathrm{nilp}})$ is a variation of mixed Hodge structures ([2] 2.3).

The construction of W is compatible with passage to the dual, as well as to tensor products. It follows that if ψ is a polarization form, then W_a and W_{-a-1} are mutually orthogonal, and ψ induces a perfect pairing $\mathrm{Gr}^W_a(V_{\mathbb{Z}})$ between $\mathrm{Gr}^W_{-a}(V)$ and $\mathrm{Gr}^W_{a-w}(V)$. Further, the induced pairings, and

$$\psi(\cdot, N^k \cdot) : \mathrm{Gr}^W_{w+k}(V) \otimes \mathrm{Gr}^W_{w+k}(V) \to \mathbb{Q}$$

have positivity properties reminiscent of those of the cohomology of a nonsingular projective variety of dimension k, with the cup-product pairing to H^{2k}, the role of N being played by the cup-product with the first Chern class of an ample line bundle.

In the terminology of [2] 1.16, $(V, W, F^{\mathrm{nilp}})$ is polarized by ψ and N. The reason for this analogy is not understood.

3 3. Warning

In general, W and the original Hodge filtration F do not define a mixed Hodge structure in a neighborhood of 0. Consider, for instance, a general pencil $f : X \to \mathbb{P}^1$ of hypersurfaces of degree d in $\mathbb{P}^M$. The fiber $X_{(\lambda,\mu)}$ at the point with projective coordinates (λ, μ) is the hypersurface with equation

$$\lambda F + \mu G = 0$$

(F, G general forms of degree d).

Let S be the set of critical values of f. On $\mathbb{P}^1 - S$, we consider the variation

$$V_{\mathbb{Z}} := R^w f_* \mathbb{Z}$$

and restrict it to a punctured disc D^* centered at a critical value (λ_0, μ_0).

We assume that

(a) M is even, hence X_s is of odd dimension $w = M - 1$.

The cohomology $H^w(X_s, \mathbb{Z})$ is torsion-free, primitive (i.e. annihilated by cup product with the hyperplane section class) and the polarization form is given by the cup product.

One knows that the global monodromy, image of $\pi_1(\mathbb{P}^1 - S)$, is Zariski dense in the corresponding symplectic group. Provided that $H^w(X_s, \mathbb{Z}) \neq 0$, i.e. $d \neq 1, 2$, N is nilpotent of rank one. The weight filtration is

$$0 \subset W_{w-1} = \mathrm{im}(N) \subset W_w = \ker(N) = V_{\mathbb{Q}}.$$

If $M \geq 4$ and d is large enough, so that $V_s \neq F^{(w-1)/2}$.

If (W, F) were a mixed Hodge structure on D^*, Gr^W_{w-1}, being one-dimensional, would have to be of type $((w-1)/2, (w-1)/2)$:

$$\mathrm{im}(N) \subset F^{(w-1)/2}.$$

By analytic continuation, any image of $\mathrm{im}(N)$ by the global monodromy group would still be in $F^{(w-1)/2}$. Those images span V, by Zariski density of the global monodromy group in the symplectic group, contradicting (b).

4 4

We now consider a case where (W, F) is a mixed Hodge structure. Fix a base point $s_0 \in D^{*n}$, let G be the group of automorphisms of $V_{\mathbb{Q}}$ respecting the form ψ, and let G_0 be the connected component of the identity of G, viewed as a linear algebraic group: it is a special orthogonal group for w even, a symplectic group for w odd.

The filtrations W, F and F^{nilp} correspond to isotropic flags:

$$(P_i)_i := F^{w-i+1}, \quad (Q_i)_i := W_{w+i-1}$$

and similarly for F^{nilp}. The subgroups of G_0 stabilizing them are parabolic subgroups.

Proposition 1 (4.1)**.** *If the parabolic subgroups $P(F^{nilp})$ and $P(W)$ defined by F^{nilp} and W satisfy*

$$\mathrm{Lie}\, P(F^{nilp}) + \mathrm{Lie}\, P(W) = \mathrm{Lie}\, G_0, \tag{4.1.1}$$

then, near 0, (W, F) is a mixed Hodge structure, with the same Hodge numbers as (W, F^{nilp}).

Remarque 2. As the $\exp(2\pi i s_j N_j) \in G_0$ respect W and permute transitively the F^{nilp} taken at various points, the validity of (4.1.1) does not depend on the point s_0 at which it is considered.

Proof. The map $u \mapsto s_0 \exp(2\pi i u)$ maps a product of upper half-planes to D^{*n}. The pullback of the local system $V_{\mathbb{Z}}$ is a trivial local system, and we can consider the Hodge filtration F at $s_0 \exp(2\pi i u)$ as a filtration $\varphi(u)$ of the fixed vector space $V_{\mathbb{C}}$.

If one wants to view $V_{\mathscr{O}}$ as a constant vector bundle as in 2.2, $\varphi(u)$ is to be replaced by $\exp(-uN)\varphi(u)$.

In particular, if we write F^{nilp} for F^{nilp} at s_0,

$$F^{\mathrm{nilp}} = \lim \exp(-uN)\varphi(u).$$

Here the limit is taken as the infimum of the $\Im(u_j) \to \infty$, i.e. $s_0 \exp(2\pi i u) \to 0$.

From (4.1.1), it then follows that for $\inf \Im(u_i)$ large, the pair of filtrations $(W, \exp(-uN)\varphi(u))$ is conjugate to (W, F^{nilp}) by $g \in G(\mathbb{C})$ close to 1. In particular,

$$\dim(W_a \cap F^{\mathrm{nilp} b}) = \dim(W_a \cap \exp(-uN)\varphi(u)^b).$$

As $\exp(-uN)\varphi(u)$ is close to F^{nilp}, this equality of dimensions implies that the filtration induced by $\exp(-uN)\varphi(u)$ on $\mathrm{Gr}^W_a(V_{\mathbb{Q}} \otimes \mathbb{C})$ tends to that induced by F^{nilp}. As N_j respects W and acts trivially on $\mathrm{Gr}^W_*(V_\mathbb{Q}\otimes\mathbb{C})$, the filtrations $\varphi(u)$ and $\exp(-uN)\varphi(u)$ induce the same filtration on $\mathrm{Gr}^W_a(V_\mathbb{Q}\otimes\mathbb{C})$.

For $\inf\Im(u_i)$ large enough, this induced filtration defines a Hodge structure of weight a, being close to the Hodge filtration $\mathrm{Gr}^W_a(F^{\mathrm{nilp}})$.

It follows that (W,F) is mixed Hodge, with the same Hodge numbers as (W,F^{nilp}). $\square$

5 5. Variant

Fix some Hodge structures H_α, and some horizontal tensors t_α, integral of type $(0,0)$, in some tensor powers

$$\bigotimes_a V_\mathbb{Z}^{\otimes n(\alpha)}\otimes\bigotimes_b V_\mathbb{Z}^{*\otimes m(\alpha)}\otimes H_\alpha.$$

One can then repeat 4.1 and its proof with G replaced by the subgroup fixing the t_α.

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We now assume that the mixed Hodge structure $(V_\mathbb{Z},W,F^{\mathrm{nilp}})$ is Hodge-Tate, i.e. such that

$$\mathrm{Gr}^W_k \text{ is of type } (-k/2,-k/2) \text{ for } k \text{ even, and } 0 \text{ for } k \text{ odd.} \tag{6.1}$$

This should be viewed as an assumption of maximal degeneracy. It means that $F_p^{\mathrm{nilp}}\oplus W_{2p-2}\to V_\mathscr{O}$ is an isomorphism for all p. If we set

$$V_{(p)} := F_p^{\mathrm{nilp}}\cap W_{2p} = F_p^{\mathrm{nilp}}\cap W_{2p+1},$$

$V_\mathscr{O}$ is then the direct sum of the $V_{(p)}$, with

$$F_p^{\mathrm{nilp}} = \bigoplus_{a\geq p} V_{(a)},\quad W_{2p}=W_{2p+1}=\bigoplus_{b\leq p} V_{(b)}.$$

The assumption (4.1.1) holds and 4.1 is in fact easier to prove in this case. In terms of the nilpotent orbit trivialization of $V_\mathscr{O}$, the filtrations W and F^{nilp} are constant. As W and F^{nilp} are opposed, and F and F^{nilp} coincide at 0, W and F remain opposed in a neighborhood of 0:

$$V_\mathscr{O} = \bigoplus V_{(a)} \quad \text{with} \quad V_{(a)} := F^p\cap W_{2p} = W_{2p+1} \tag{6.2}$$

implying that $(V_\mathbb{Z},W,F)$ is mixed Hodge and Hodge-Tate. We will assume D^n has been shrunk so that $(V_\mathbb{Z},W,F)$ is mixed Hodge on the whole of D^{*n}.

Our aim is to classify, i.e. to give another description, of variations such as the above.

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In the category of mixed Hodge structures, $\mathrm{Hom}(\mathbb{Z}(0),\mathbb{Z}(1))=0$ (notations: see 1): an extension of $\mathbb{Z}(0)$ by $\mathbb{Z}(1)$ has no nontrivial automorphism as extension. The extensions are classified by $\mathbb{C}^*$:

$$\mathrm{Ext}^1(\mathbb{Z}(0),\mathbb{Z}(1))=\mathbb{C}^*, \tag{7.1}$$

with $q\in\mathbb{C}^*$ corresponding to the extension $0\to\mathbb{Z}(1)\to H_q\to\mathbb{Z}(0)\to 0$.

$$\begin{aligned} H_\mathbb{C}&\cong\mathbb{C}^2,\ \text{basis } e_0,e_1;\\ W_{-2}&=\mathbb{C}e_1,\quad F^0=\mathbb{C}e_0;\\ H_\mathbb{Z}&=2\pi i\mathbb{Z}\cdot e_1+\mathbb{Z}\cdot(e_0+\log q\cdot e_1)\subset H_\mathbb{C};\\ \alpha(2\pi i)&=2\pi i e_1,\quad \beta(e_0)=1.\end{aligned} \tag{7.2}$$

Note that the choice of the determination of log does not matter.

Equivalent description: H_q has a basis f_0, f_1 with $\alpha(2\pi i) = f_0$, $\beta(f_1) = 1$, $F^0 \subset H_{\mathbb{C}}$ the kernel of $\alpha : H_{\mathbb{C}} \to \mathbb{C}$.

Variations of mixed Hodge structures H over S that are extensions of $\mathbb{Z}(0)$ by $\mathbb{Z}(1)$ are similarly classified by invertible holomorphic functions. The extension H defined by q has $H_{\mathscr{O}} = \mathscr{O} \cdot e_0 \oplus \mathscr{O} \cdot e_1$, with basis e_0 (spanning F^0) and e_1 (spanning W_{-2}) and integral lattice and structure maps α, β as in (7.2). The connection is given by

$$\nabla = d - \frac{1}{2\pi i}\frac{dq}{q}\beta. \tag{7.3}$$

We now take $S = D^{*n}$. Let n_j be the order of q along $s_j = 0$:

$$\frac{1}{2\pi i}\frac{(\log)q}{s_j} = i\partial_j \log q + n_j \cdot 2\pi i.$$

The constant sections of $H_{\mathscr{O}}$, in the sense of 2.2, are spanned by e_1 and $e_0 + (\log(q/\prod s_j^{n_j})) \cdot e_1$.

It follows that F extends to the canonical extension of $H_{\mathscr{O}}$, remaining a supplement to W_{-2}, if and only if q belongs to the group $\mathscr{O}^*_{\mathrm{mer}}(D^n)$ of invertible holomorphic functions on D^{*n}, meromorphic along the $s_j = 0$. The filtration F is constant with respect to the trivialization of $H_{\mathscr{O}}$ given in 2.2 if and only if q is a monomial $a \cdot \prod s_j^{n_j}$.

The nilpotent orbit attached to H (cf. 1.3) is again an extension of $\mathbb{Z}(0)$ by $\mathbb{Z}(1)$. Its invariant is the monomial q_0 such that q/q_0 is holomorphic invertible on D^n with value 1 at 0.

The definitions given do not require the choice of which i is i and which is $-i$ in $\mathbb{C}$. If one is willing to make that choice, one could rather say that the extensions of $\mathbb{Z}$ (type $(0,0)$) by $\mathbb{Z}$ (type $(-1,-1)$) are parametrized by $\mathbb{C}^*$. Formulas (7.2)(7.3) should be changed as follows:

$$\begin{aligned} H_{\mathbb{Z}} &= \mathbb{Z}e_1 + \mathbb{Z}(e_0 + \frac{\log q}{2\pi i}) \subset H_{\mathbb{C}}; \\ \alpha(1) &= e_1, \quad \beta(e_0) = 1; \end{aligned} \tag{7.2$'$}$$

replace $\frac{1}{2\pi i}\frac{dq}{q}$ by $\frac{1}{2\pi i}\frac{dq}{q}$ in (7.3).

More generally, for A and B free abelian groups, viewed as purely of type (p,p) and $(p-1, p-1)$, one has for the corresponding constant local system on S

$$\mathrm{Ext}^1_{\text{mixed Hodge}}(A, B) = \mathrm{Hom}(A, B) \otimes \mathscr{O}^*(S). \tag{7.4}$$

The underlying vector bundle with connection $(H_{\mathscr{O}}, \nabla)$ of the extension with class ξ is

$$\begin{aligned} H_{\mathscr{O}} &= A \otimes \mathscr{O} \oplus B \otimes \mathscr{O}, \\ \nabla &= d - \frac{1}{2\pi i} d\log \xi \cdot \beta. \end{aligned} \tag{7.5}$$

For $S = D^{*n}$, the filtration F extends to the canonical extension, remaining opposite to W at 0, if and only if

$$\xi \in \mathrm{Hom}(A, B) \otimes \mathscr{O}^*_{\mathrm{mer}}(D^n).$$

The group $\mathscr{O}^*_{\mathrm{mer}}(D^n)$ contains the group of monomials $\{a \cdot \prod s_j^{n_j}\}$, and retracts to it by a unique homomorphism nilp such that $f/\mathrm{nilp}(f)$ is holomorphic at zero, where it takes the value 1.

Let F^{nilp} be the constant filtration (2.2) agreeing with F at 0. The nilpotent orbit $(H_{\mathbb{Z}}, F^{\mathrm{nilp}})$ is again an extension of A by B. Its invariant is $\mathrm{nilp}(\xi)$, in $\mathrm{Hom}(A, B) \otimes \langle\text{monomials}\rangle$.

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With §§6 as the motivating example, we now consider on D^{*n} variations of mixed Hodge structures $(V_{\mathbb{Z}}, W, F)$ of Hodge-Tate type (6.1) such that (8.1), (8.2) below hold.

8.1

$\mathrm{Gr}^W_{2k}(V_\mathbb{Z})$ is a constant local system.

In (8.1), W denotes the filtration of $V_\mathbb{Z}$ induced by the filtration W of $V_\mathbb{Q}$. We will identify the constant local system $\mathrm{Gr}^W_{2k}(V_\mathbb{Z})$ with its constant value: a free abelian group. The conditions (8.1) imply that the monodromy is unipotent.

8.2

F extends to the canonical extension of $V_\mathscr{O}$, and, at 0, the filtrations F and W are opposed.

By (8.1)(8.2), the quotient $W_{2k}(V_\mathbb{Z})/W_{2k-4}(V_\mathbb{Z})$ is an extension of $\mathrm{Gr}^W_{2k}(V_\mathbb{Z})$ (type (k,k)) by $\mathrm{Gr}^W_{2k-2}(V_\mathbb{Z})$ (type $(k-1,k-1)$) of the type considered in §7. It has an extension class

$$B_k \in \mathrm{Hom}(\mathrm{Gr}^W_{2k}(V_\mathbb{Z}), \mathrm{Gr}^W_{2k-2}(V_\mathbb{Z})) \otimes \mathscr{O}^*_{\mathrm{mer}}(D^n).$$

Let E be the sum of those classes:

$$E \in \mathrm{End}(\mathrm{Gr}^W(V_\mathbb{Z}))_{-2} \otimes \mathscr{O}^*_{\mathrm{mer}}(D^n). \tag{8.3}$$

As in (6.1), the weight and Hodge filtration of $V_\mathscr{O}$ define a direct sum decomposition $V_\mathscr{O} = \bigoplus V^{(a)}$. One has

$$V^{(p)} \xrightarrow{\sim} \mathrm{Gr}^W_{2p}(V_\mathbb{Z}) \otimes \mathscr{O},$$

giving

$$V_\mathscr{O} \cong \mathrm{Gr}^W(V_\mathbb{Z}) \otimes \mathscr{O}. \tag{8.4}$$

Lemme 3 (9)**.** *The isomorphism (8.4) transforms the connection ∇ of $V_\mathscr{O}$ into the connection*

$$\nabla = d - \frac{1}{2\pi i} d\log E.$$

Proof. The connection ∇ of $V_\mathscr{O}$ respects W, induces the trivial connection on $\mathrm{Gr}^W(V_\mathbb{Z}) \otimes \mathscr{O}$, and maps F^p to F^{p-1}. It is hence given by a 1-form with values in endomorphisms of degree -2 of $\mathrm{Gr}^W(V_\mathbb{Z}) \otimes \mathscr{O}$. To compute it, it suffices to compute the induced connections on the quotients W_{2k}/W_{2k-4}, and we apply (7.5). □

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If we replace F by F^{nilp}, the constant filtration (2.2) agreeing with F at 0, one obtains the nilpotent orbit $(V_\mathbb{Z}, W, F^{\mathrm{nilp}})$ attached to $(V_\mathbb{Z}, W, F)$. By §§7, the corresponding invariant $E((V_\mathbb{Z}, W, F^{\mathrm{nilp}}))$ is deduced from $E = E((V_\mathbb{Z}, W, F))$ by applying

$$\mathrm{nilp} : \mathscr{O}^*_{\mathrm{mer}}(D^n) \to \{\text{monomials}\}. \tag{10.1}$$

The group of monomials is an extension of $\mathbb{Z}^n$ (the exponents) by $\mathbb{C}^*$. If one projects further to $\mathbb{Z}^n$, the information retained is the $\mathrm{Gr}^W(1-T_j) = \mathrm{Gr}^W(N_j)$: by reduction to the case of extensions of $\mathbb{Z}$ by $\mathbb{Z}(1)$, one checks that, applying ν_j, the order along $s_j = 0$, one has

$$\mathrm{Gr}^W(1-T_j) = \mathrm{Gr}^W(N_j) = -\nu_j(E) \in \mathrm{End}(\mathrm{Gr}^W(V_\mathbb{Z}))_{-2}. \tag{10.2}$$

The integrability of ∇ can be written

$$d\log E \wedge d\log E = 0 \quad \text{in} \quad \mathrm{End}(\mathrm{Gr}^W) \otimes \Lambda^2 \otimes \mathbb{Q}. \tag{10.3}$$

For $(V_\mathbb{Z}, W, F^{\mathrm{nilp}})$, i.e. for the dominant part E_0, it corresponds to the commutativity of the $\mathrm{Gr}^W(N_i)$.

Let us consider pairs $((V_{\mathbb{Z}}, W, F^{\mathrm{nilp}}), E)$ of a mixed Tate nilpotent orbit $(V_{\mathbb{Z}}, W, F^{\mathrm{nilp}})$, whose invariant will be denoted E_{nilp}, and of

$$E \in \mathrm{End}(\mathrm{Gr}^W(V_{\mathbb{Z}}))_{-2} \otimes \mathscr{O}^*_{\mathrm{mer}}(D^n)$$

such that

$$d\log E \wedge d\log E = 0 \quad \text{and} \quad E_{\mathrm{nilp}} = \mathrm{nilp}(E).$$

We turn those pairs into a category by defining a morphism f from $((V_{\mathbb{Z}}, W, F^{\mathrm{nilp}}), E)$ to $((V'_{\mathbb{Z}}, W', F^{\mathrm{nilp}\prime}), E')$ to be $f : V_{\mathbb{Z}} \to V'_{\mathbb{Z}}$, compatible with W and F^{nilp}, and such that $\mathrm{Gr}^W(f) \circ E = E' \circ \mathrm{Gr}^W(f)$.

Théorème 4 (11)**.** *To $(V_{\mathbb{Z}}, W, F)$ as in §§8, let us attach*

(a) the corresponding nilpotent orbit $(V_{\mathbb{Z}}, W, F^{nilp})$,

(b) $E \in \mathrm{End}(\mathrm{Gr}^W(V_{\mathbb{Z}}))_{-2} \otimes \mathscr{O}^*_{mer}(D^n)$.

This construction is an equivalence from the category of variations $(V_{\mathbb{Z}}, W, F)$ as in §§8 to the category of pairs as above.

Remarque 5. The functor of the theorem is compatible with tensor products, with the E of $V \otimes V'$ defined to be $E \otimes \mathrm{Id}_{V'} + \mathrm{Id}_V \otimes E'$. It is also compatible with duals (with E going to the opposite of its transpose) and, combining the two, with $\underline{\mathrm{Hom}}$.

Proof. A morphism of V to V' is a section of $\mathrm{Hom}(V, V')_{\mathbb{Z}}$ in W_0 and F^0. Thus, to show that the function is fully faithful, it suffices to show that a global section v of $V_{\mathbb{Z}}$ is in W_0 and F^0 if it is in W_0, in F_0^{nilp} and if its image $v^{(0)}$ in $\mathrm{Gr}_0^W(V_{\mathbb{Z}})$ is killed by E: $Ev^{(0)}$ trivial in $\mathrm{Gr}_{-2}^W(V_{\mathbb{Z}}) \otimes \mathscr{O}^*$.

Being global and horizontal, v is constant, in the sense of (2.2). Let $v^{(0)}$ be its component in $V^{(0)}$ (6.2). Via the isomorphism of $V_{\mathscr{O}}^{(0)}$ with $\mathrm{Gr}_0^W(V_{\mathbb{Z}}) \otimes \mathscr{O}$ (8.4), it corresponds to $\bar{v}$ in $\mathrm{Gr}_0^W(V_{\mathbb{Z}})$. By Lemma 9, as $\xi \cdot \bar{v}$ is trivial, $v^{(0)}$ is horizontal. The two constant sections v and $v^{(0)}$ coincide at 0, as v is in F^{nilp}. It follows that they are equal, and $v = v^{(0)} \in F^0$.

It remains to show that any $((V_{\mathbb{Z}}^{\mathrm{nilp}}, W^{\mathrm{nilp}}, F^{\mathrm{nilp}}), E)$ is the image by the functor of some $(V_{\mathbb{Z}}, W, F)$. The sought after $V_{\mathscr{O}}$, with its weight and Hodge filtration, and connection, is

$$V_{\mathscr{O}} = \mathrm{Gr}^W(V_{\mathbb{Z}}^{\mathrm{nilp}}) \otimes \mathscr{O}.$$

Its canonical extension is $\mathrm{Gr}^W(V_{\mathbb{Z}}^{\mathrm{nilp}}) \otimes \mathscr{O}$ on D^n, and the constantification (2.2) gives

$$\phi : \mathrm{Gr}^W(V_{\mathbb{Z}}^{\mathrm{nilp}}) \otimes \mathscr{O} \to V_{\mathscr{O}}$$

with $\mathrm{Gr}^W(V_{\mathbb{Z}}^{\mathrm{nilp}})$ mapping to constant sections. The map ϕ is horizontal if, at the source, one uses

$$\nabla = d - \frac{1}{2\pi i} d\log E_{\mathrm{nilp}}.$$

The vector bundle with connection $\mathrm{Gr}^W(V_{\mathbb{Z}}^{\mathrm{nilp}}) \otimes \mathscr{O}, \nabla_0$ is the one underlying $(V_{\mathbb{Q}}^{\mathrm{nilp}}, W^{\mathrm{nilp}}, F^{\mathrm{nilp}})$.

The sought after $(V_{\mathbb{Z}}, W, F)$ is given by

$$V_{\mathbb{Z}} = \phi(V_{\mathbb{Z}}^{\mathrm{nilp}}) \subset V_{\mathscr{O}}.$$

Applying the functor, it gives back $(V_{\mathbb{Q}}^{\mathrm{nilp}}, W^{\mathrm{nilp}}, F^{\mathrm{nilp}})$. To check that it gives back E, it suffices to check it on the quotients W_{2k}/W_{2k-4}. This case in turn reduces to that of extensions of $\mathbb{Z}$ by $\mathbb{Z}(1)$, which is left to the reader. □

10 12

We now apply Theorem 11 in the situation of §§6. To a polarized variation V on D^{*n}, with (V, W, F^{nilp}) of Hodge-Tate type, we attach the corresponding polarized nilpotent orbit (V, F^{nilp}) and $E \in \text{End}(\text{Gr}^W(V_{\mathbb{Z}}))_{-2} \otimes \mathscr{O}^*_{\text{mer}}(D^n)$.

The fact that ψ is a morphism $V \otimes V \to \mathbb{Z}(-w)$, and hence induces a morphism $(V_{\mathbb{Z}}, W, F) \otimes (V_{\mathbb{Z}}, W, F) \to \mathbb{Z}(-w)$, translates into

$$\psi(Ex, y) + \psi(x, Ey) = 0 \quad \text{in} \quad \mathscr{O}^*_{\text{mer}}(D^n) \tag{12.1}$$

for $x \in \text{Gr}^W_k(V_{\mathbb{Z}})$, $y \in \text{Gr}^W_{k'}(V_{\mathbb{Z}})$ and $k + k' = 2$.

Conversely, given

(a) a polarized nilpotent orbit (V, F^{nilp}) with (V, W, F^{nilp}) of Hodge-Tate type,

(b) $E \in \text{End}(\text{Gr}^W(V_{\mathbb{Z}}))_{-2} \otimes \mathscr{O}^*_{\text{mer}}(D^n)$ with $d \log E \wedge d \log E = 0$, E compatible with E_0 attached to (V, W, F^{nilp}): $E - E_0$ holomorphically trivial at 0, and E obeying (12.1),

we obtain by 11 a mixed variation (V, W, F). By (12.1), $\psi : (V_{\mathbb{Z}}, W, F) \otimes (V_{\mathbb{Z}}, W, F) \to \mathbb{Z}(-w)$ is a morphism. If we forget W, $(V_{\mathbb{Z}}, F)$ admits (V, F^{nilp}) as nilpotent orbit. By 2.3, $(V_{\mathbb{Z}}, F)$ is a polarized variation near 0.

11 13. Example

Let E be a family of elliptic curves parametrized by D^*, and consider the variation $H_1(E_s)$ ($s \in D^*$). Assume the monodromy is unipotent and nontrivial. In that case, one can uniquely write

$$E_s = \mathbb{C}^*/q^{\mathbb{Z}}.$$

with $q(s)$ holomorphic invertible on D^* and tending to 0 with s. We are in the situation of (4.1) and the mixed variation $(H_1(E_s), W, F)$ is the extension of $\mathbb{Z}(0)$ by $\mathbb{Z}(1)$ with invariant $q(s)$.

In this case, the mixed Hodge structures $(H_1(E_s), W, F)$ are subquotients of the homology of algebraic varieties: they are "motivic". Indeed, the extension of $\mathbb{Z}(0)$ by $\mathbb{Z}(1)$ with invariant q ($q \neq 1$) is the homology (H_1) of $\mathbb{G}_m(\mathbb{C}) - \{1, q\}$ with the points 0 and ∞ identified. To check this, apply (P. Deligne, Théorie de Hodge III, Publ. Math. IHES 44 (1974), pp. 5–78, construction 10.3.8).

More generally, for a family of polarized abelian varieties A on D^{*n}, and their H_1, 4.1 applies and the resulting mixed Hodge structures are motivic.

I do not expect this to hold in general. Let V be a direct factor of $H^w(X_s)$, for X a family of projective nonsingular varieties over D^{*n}. The nilpotent orbit (V, W, F^{nilp}) should be motivic but, even if (V, W, F^{nilp}) is Hodge-Tate, the $(V_{\mathbb{Z}}, W, F)$ should not always be. If they were, it would conflict with another conjecture: for an iterated extension E of $\mathbb{Z}(0)$ by $\mathbb{Z}(1)^n$ by $\mathbb{Z}(2)$, defining extension classes

$$q_{1i} \in \text{Ext}^1(\mathbb{Z}(0), \mathbb{Z}(1)^n) = \mathbb{C}^{*n}$$

$$q_{2j} \in \text{Ext}^1(\mathbb{Z}(1)^n, \mathbb{Z}(2)) = \mathbb{C}^{*n},$$

if E is motivic, then $\prod_{i,j} \{q_{1i}, q_{2j}\} \in K_2(\mathbb{Q})$ is trivial. As a consequence,

$$q_{1i} \frac{d}{dq_{1i}} \wedge q_{2j} \frac{d}{dq_{2j}} = 0$$

in $\wedge^2 \mathbb{Q}^*$ and, in the case $n = 1$, q_1 and q_2 are algebraically dependent over $\mathbb{Q}$.

Let us consider the family of quintic threefolds

$$\sum_{i=1}^{5} X_i^5 - \psi \prod_{i=1}^{5} X_i = 0.$$

Let G be the multiplicative group of 5-uples of fifth roots of 1 with $\prod \zeta_i = 1$. It acts on each quintic in the family: $(X_i) \mapsto (\zeta_i X_i)$, with the quotient Γ of G by the diagonal (all ζ_i equal) acting faithfully. Let V be the part of H^3 fixed by G. By the computations of [1], V is of rank 4 and the monodromy group is Zariski dense in Sp_4: it contains a unipotent transformation u with $u-1$ of rank one (monodromy around $\lambda = \zeta$, for $\zeta^5 = 1$), and one with $u-1$ of rank 3, i.e. one Jordan block (monodromy around $\psi = \infty$). As the Zariski closure of the monodromy is known to be semi-simple, and contained in Sp_4, this leaves no choice.

Near $\psi = \infty$, the nilpotent orbit is of Hodge-Tate type, so that E gives rise to an iterated extension of $\mathbb{Z}(-3)$ by $\mathbb{Z}(-2)$ by $\mathbb{Z}(-1)$ by $\mathbb{Z}(0)$. Let q_1 (resp. q_2) be the class of the resulting extension of $\mathbb{Z}(-3)$ by $\mathbb{Z}(-2)$ (resp. of $\mathbb{Z}(-2)$ by $\mathbb{Z}(-1)$). If, for every value of λ near ∞, q_1 and q_2 are algebraically dependent over $\mathbb{Q}$, a Baire category argument and analytic continuation show that $F(q_1, q_2) = 0$ identically for some polynomial F.

For all but finitely many points (q_1^0, q_2^0) on the algebraic curve $F(q_1, q_2) = 0$, in a neighborhood of (q_1^0, q_2^0), the equation $F(q_1, q_2) = 0$ amounts to $q_2/q_2^0 = \varphi(q_1/q_1^0)$ for some function φ depending on (q_1^0, q_2^0). We write $\varphi_\infty[q_1^0, q_2^0]$ for that function. It maps 1 to 1, and its ∞-jet at 1, $\varphi_{\infty\infty}[q_1^0, q_2^0]$, depends algebraically on (q_1^0, q_2^0), i.e. for each k, the k-jet $\varphi_{\infty k}[q_1^0, q_2^0]$ does so.

The ∞-jet $\varphi_{\infty\infty}[q_1^0, q_2^0]$ carries the following information. Fix λ_0 near ∞, giving rise to (q_1^0, q_2^0). Fix on V a new integral basis e_i $(0 \leq i \leq 3)$, with e_i in $F^{-i} \cap W_{-2i}$ and projecting in $\mathrm{Gr}^W_{-2i}(V) \cong \mathbb{Z}(-i)$ to the integral generator. In this new basis $V_\mathbb{Z}$ is the split mixed Hodge structure $\bigoplus_0^3 \mathbb{Z}(-i)$. The ∞-jet $\varphi_{\infty\infty}[q_1^0, q_2^0]$ tells how, in this basis, the Hodge filtration changes with λ, up to a reparametrization $\lambda \mapsto g(\lambda)$.

Let $\mathbb{P}(V)$ be the projective space of rays in V. The symplectic form ψ gives on $\mathbb{P}(V)$ a "null-system" of lines, corresponding to the Lagrangian subspaces of V. The variable Hodge filtration F gives a curve Γ in $\mathbb{P}(V)$: the trajectory of F^3. At a general point of Γ, the tangent line is a null line, corresponding to F^2. The weight filtration gives an isotropic flag. The previous discussion gives: for each $\gamma \in \Gamma$, let us consider the relative position of the ∞ order jet R_γ of Γ at γ, and of ψ. Here, "relative position" is an element of the quotient by $\mathrm{Sp}(4)$ of the space of pairs (∞-order jet of curve, isotropic flag). For variable γ, this relative position is controlled by a $\varphi_{\infty\infty}[q_1^0, q_2^0]$, hence stays on a 1-dimensional algebraic subspace. If we analytically continue and come back to γ, we see that (R_γ, ψ') is again in this subspace, for a Zariski dense set of ψ', hence for all ψ'. As the space of all isotropic flags is 4-dimensional, there is a 3-dimensional space of ψ' such that all (R_γ, ψ') are in the same relative position.

The ∞-jet R_γ is hence stable by a subgroup H of $\mathrm{Sp}(4)$, fixing γ, and of dimension ≥ 3. The curve Γ spans $\mathbb{P}(V)$, hence cannot be pointwise fixed. It must be an orbit of H, hence is an algebraic curve. Contradiction: no algebraic curve has a group of automorphisms of dimension ≥ 3, all fixing a point γ.

12 14. Example

The following occurs in the study of mirror symmetry.

Let V be a polarized variation of Hodge structure on D^{*n}, of the following type:

(14.1) weight -3 (like an H^3), Hodge numbers $h^{p,q}$ $(p = 0, -1, -2, -3)$: $1, n, n, 1$.

(14.2) polarization form ψ of discriminant 1.

(14.3) unipotent monodromy, and associated nilpotent orbit of Hodge-Tate type.

Rank of the $\mathrm{Gr}_i^W(V_\mathbb{Z})$ ($i = 0, -2, -4, -6$): $1, n, n, 1$. The form ψ is a perfect duality between Gr_0^W and Gr_{-6}^W.

(14.4) For 1 a basis of $\mathrm{Gr}_0^W(V_\mathbb{Z})$, the $N_i(1)$ form a basis of $\mathrm{Gr}_{-2}^W(V_\mathbb{Z})$.

Fix a basis $1 \in \mathrm{Gr}_0^W(V_\mathbb{Z})$. By definition of the weight filtration and (14.4), $\mathrm{Gr}^W(V_\mathbb{Q})$, as a module over the polynomial algebra $\mathbb{Q}[N_1, \ldots, N_n]$, is generated by 1. As a quotient of $\mathbb{Q}[N_1, \ldots, N_n]$, it inherits an algebra structure, with unit 1 and with $N_i =$ multiplication by $N_i(1)$. The integral graded module $\mathrm{Gr}^W(V_\mathbb{Z})$ is a subalgebra: as $\mathrm{Gr}^W(V_\mathbb{Z}) = 0$ for $i \neq 0, -2, -4, -6$, it suffices to check that it is stable by multiplication by $v \in \mathrm{Gr}_{-2}^W(V_\mathbb{Z})$, $i = 0$ or -2 which is clear: $\mathrm{Gr}_0^W(V_\mathbb{Z}) = \mathbb{Z} \cdot 1$ and, if L is the free module spanned by $N_1, \ldots, N_n$,

$$L \xrightarrow{\sim} \mathrm{Gr}_{-2}^W(V_\mathbb{Z}) : \ell \mapsto \ell \cdot 1.$$

The form induced by ψ can be recovered from the linear form t on $\mathrm{Gr}_{-6}^W(V_\mathbb{Z})$:

$$t(v) := \psi(1, v). \text{ Indeed, for } \ell \in \mathrm{Gr}_{-2}^W(V_\mathbb{Z}), \lambda \in \mathrm{Gr}_{-4}^W(V_\mathbb{Z}), \text{ one has}$$

$$\psi(\ell \cdot 1, \lambda) = -\psi(1, \ell \cdot \lambda) = t(\ell \cdot \lambda).$$

The algebra structure, and t, can be recovered from the symmetric trilinear form

$$\Phi(x, y, z) := t(x \cdot y \cdot z)$$

on $L = \mathrm{Gr}_{-2}^\mathbb{Q}(V_\mathbb{Z})$: the algebra is $\mathbb{Z} \oplus L \oplus L^\vee \oplus \mathbb{Z}$, the first $\mathbb{Z}$ spanned by 1, the second by $1^\vee$ with $t(1^\vee) = 1$. The product of $\lambda \in L^\vee$ with $\lambda' \in L^\vee$ is $\lambda(\lambda') \cdot 1^\vee$, while that of $x, y \in L$ is the linear form $z \mapsto \Phi(x, y, z)$.

We now consider the extension classes

$$E_0 \in \mathrm{Ext}^1(\mathrm{Gr}_0^W(V_\mathbb{Z}), \mathrm{Gr}_{-2}^W(V_\mathbb{Z})) = L \otimes \mathscr{O}^*_{\mathrm{mer}}(D^n) = \mathrm{Hom}(L^\vee, \mathscr{O}^*_{\mathrm{mer}}(D^n))$$
$$E_1 \in \mathrm{Ext}^1(\mathrm{Gr}_{-2}^W(V_\mathbb{Z}), \mathrm{Gr}_{-4}^W(V_\mathbb{Z})) = L^\vee \otimes L^\vee \otimes \mathscr{O}^*_{\mathrm{mer}}(D^n) = \mathrm{Hom}(L \otimes L, \mathscr{O}^*_{\mathrm{mer}}(D^n))$$
$$E_2 \in \mathrm{Ext}^1(\mathrm{Gr}_{-4}^W(V_\mathbb{Z}), \mathrm{Gr}_{-6}^W(V_\mathbb{Z})) = L \otimes \mathscr{O}^*_{\mathrm{mer}}(D^n) = \mathrm{Hom}(L^\vee, \mathscr{O}^*_{\mathrm{mer}}(D^n)).$$

The valuation of the E_k along $s_j = 0$ are given by N_j (10.2). For E_0, recalling that $L = \mathbb{Z}^n$, we find

$$E_0 : (q^1, \ldots, q^n)$$

and the q^i form a system of local coordinates near 0. For $\lambda = (a_1, \ldots, a_n) \in L^\vee \cong \mathbb{Z}^n$, we write $q^\lambda = \prod q_i^{a_i}$. We have then

$$E_0 : \lambda \longmapsto q^\lambda,$$
$$E_1 : \ell \otimes m \longmapsto q^{\Phi(\ell,m)} \cdot U_{\ell,m}, \quad \text{with } U_{\ell,m} \text{ invertible on } D^n,$$
$$E_2 : \lambda \longmapsto q^\lambda;$$

compatibility with ψ imposes $E_2(\lambda) = E_1(\lambda)$, $E_1(\ell, m) = E_1(m, \ell)$.

Let us expand the $U_{\ell,m}$ as an infinite product, extended over the $\lambda \in L^\vee$, $\lambda = (a_1, \ldots, a_n)$, with $a_i \geq 0$, $\sum a_i \neq 0$:

$$U_{\ell,m} = C_{\ell,m} \prod_\lambda (1 - q^\lambda)^{c(\ell,m,\lambda)}$$

with $C_{\ell,m}$ a constant, bi-multiplicative in ℓ, m and $c(\ell, m, \lambda)$ complex, bi-additive in ℓ, m. The integrability condition 10.3 reduces to

$$c(\ell_1, m, \ell_2, \lambda)/c(\ell_2, m, \ell_1, \lambda) = c(\ell_1, \ell_2, m, \lambda)/c(\ell_2, \ell_1, m, \lambda).$$

Thus we obtain an expansion

$$E_1(\ell, m) = C_{\ell,m} \cdot q^{\Phi(\ell,m)} \prod_\lambda (1 - q^\lambda)^{\lambda(\ell)\lambda(m)\zeta(\lambda)}$$

In the mirror story, $\mathbb{Z}$, L, $L^\vee$, $\mathbb{Z}$ become the H^0, H^2, H^4, H^6 of a Calabi–Yau threefold, the basis of L some basis of H^2 contained in the ample cone, and λ runs over the cohomology class of rational curves (in H_4), with $\zeta(\lambda)$, the number of rational curves in a given class, suitably defined.

For the components of E_1 in the given basis of $L = \mathbb{Z}^n$, and the dual basis of $L^\vee$, this gives

$$\partial_i\partial_j\partial_k \log E_1(e_i, e_j) = \Phi(e_i, e_j, e_k) + \sum_\lambda \zeta(\lambda)\lambda(e_i)\lambda(e_j)\lambda(e_k)\frac{q^\lambda}{1-q^\lambda}$$

having as logarithmic derivative

$$\Phi(e_i, e_j, e_k) + \sum_{\lambda\neq 0} \zeta(\lambda)\lambda(e_i)\lambda(e_j)\lambda(e_k)q^\lambda.$$

13 15. Integrality

For the degenerating elliptic curve 13 over D^* (coordinate z), the formal power series $\sum a_n z^n$, Taylor expansion of q as a function of z, has a purely algebraic sense: if the coefficients c_α, of an equation for E are expanded as formal power series in z: $c_\alpha = \sum c_{\alpha n} z^n$, then the a_n are polynomials in the $c_{\alpha m}$.

Further, the story makes sense in any characteristic, and is compatible with reduction modulo p. I expect similar integrality to hold in general.

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